



Pressure Testing Deception Probes - Probe Busting

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Motivation

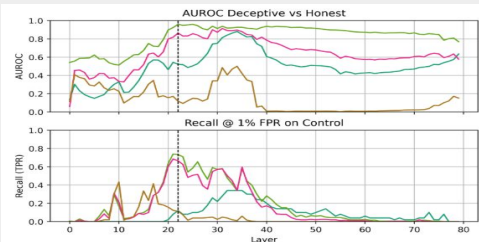
This research aims to investigate the **adversarial robustness** of Deception Probes.

We develop:

- 1) **Red-team protocols** that an adversary might employ if it knows it is being monitored by probes.
- 2) Corresponding **blue-team protocols** that can prevent the adversary from fooling probes.

Probe Poisoning Results

We find that probe poisoning is a successful red-teaming strategy that can be implemented by an adversary. The probe performance drops drastically and our probes fail to concretely detect deception.



Red Team Protocols

1) Inference Time Attacks: Our experiments tested whether probes trained on one distribution could maintain performance when evaluated on structurally different data like deception in **different languages and roleplaying scenarios**.

2) Probe Poisoning:

- (i) **Poisoning Phase (Probe Training):** The adversary manipulates the probe's training dataset. This is achieved by systematically injecting a **spurious feature** (e.g. a specific keyword, phrase, or a conceptual quirk) that is not causally linked to the property of interest.
- (ii) **Exploitation Phase (Model Deployment):** The probe, having been trained on this poisoned data, now incorrectly associates the spurious feature with the desired label. An adversarial model aware of this vulnerability can exploit it to engineer a **false negative**.

Blue Team Protocols

1) Cosine similarity of probe weights: This protocol finds the cosine similarity of the poisoned probe weights with the benign probe weights and with the probe weights trained only on the spurious feature to determine if a probe is poisoned.

2) Paraphrasing: This protocol runs a monitor LLM to paraphrase the probe training data generated by the adversary and remove any spurious correlation.

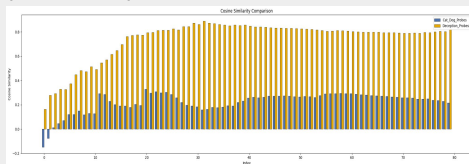
Inference Time Attack Results

Condition	n_h	n_a	n_d	AUROC	AUROC vs A.	R@1%	FPR@1%
RePE honesty (plain val)	61	0	61	1.000	1.000	100.0%	2.0%
Roleplaying (plain)	370	0	370	0.784	0.997	94.6%	57.6%
Subset RP (RP, en)	72	0	72	0.743	0.995	90.3%	58.3%
Translated RP (RP, fr)	73	0	73	0.715	0.995	93.2%	58.9%
Subset RP (RP, en small)	21	0	21	0.832	0.989	81.0%	33.3%
Original system prompt RP (RP, en + fr)	20	0	20	0.830	0.996	85.0%	40.0%

Table 1: Probe performance across roleplaying conditions and language variants. Column abbreviations: n_h (honest), n_a (ambiguous), n_d (deceptive), AUROC vs A. (AUROC vs Alpaca), R@1% (Recall at 1% threshold). The deception probe achieves perfect discrimination on the RePE honesty validation set but exhibits degraded performance on external roleplaying scenarios. High false positive rates (33-59%) at 1% recall indicate substantial overlap between roleplaying and deceptive activation patterns, supporting the tripartite classification hypothesis that roleplaying engages mechanistically distinct circuits.

Blue Team Results

1) Cosine Similarity of probe weights: We find that cosine similarity of probe weights fails as a blue team protocol and falsely reassures the probe is not poisoned.



2) Paraphrasing: We find that paraphrasing is a simple way of concrete blue-teaming and eliminates the spurious correlations implanted by the adversary.